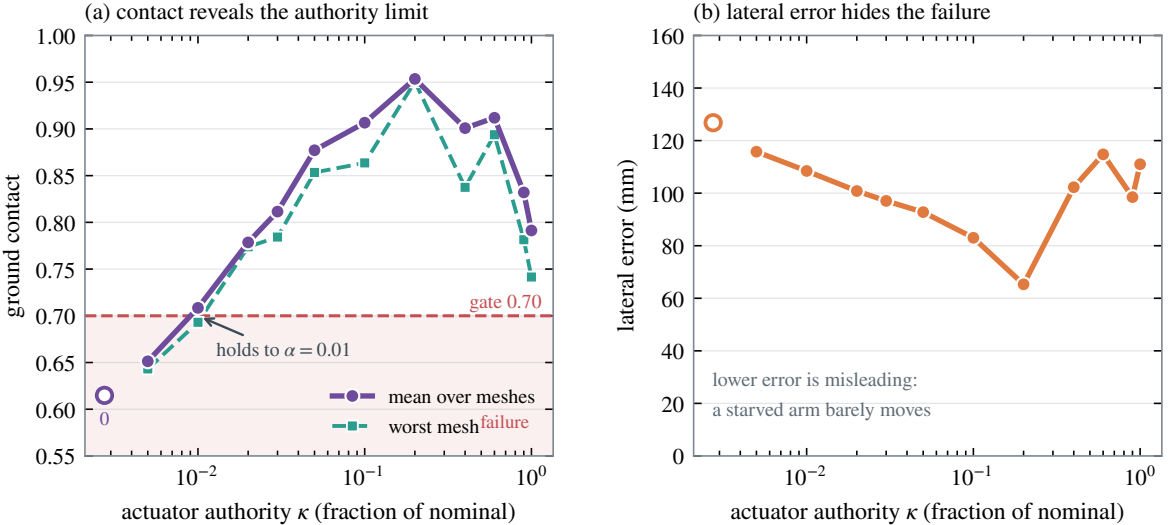
(a) contact reveals the authority limit
ground contact
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
gate 0.70
holds to $\alpha = 0.01$
0
mean over meshes
worst mesh
failure
10^{-2}
10^{-1}
10^{0}
actuator authority κ (fraction of nominal)
(b) lateral error hides the failure
lateral error (mm)
160
140
120
100
80
60
40
20
0
lower error is misleading:
a starved arm barely moves
10^{-2}
10^{-1}
10^{0}
actuator authority κ (fraction of nominal)